

An Evaluation of MedGemma for Anterior Segment Eye Images Classification and Analysis

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Abstract—Cataract grading via Lens Opacities Classification System III (LOCS III) is clinically essential yet prone to inter-observer variability. This study evaluates MedGemma, a biomedical Visual-Language Model (VLM), using zero-shot inference on anterior segment images to predict LOCS III grades and generate clinical prose, benchmarked against expert annotations. To address the *Evaluation Gap*, we implemented a multidimensional framework of lexical (Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence (ROUGE-L)) and semantic (Bidirectional Encoder Representations from Transformers Score (BERTScore), Sentence-BERT (sBERT)) metrics, featuring a cross-validation baseline of mutually exclusive diagnoses. While quantitative results indicated high semantic similarity (BERTScore: 0.845, sBERT: 0.788) despite low lexical overlap (ROUGE-L: 0.213), our baseline proved that semantic metrics artificially inflate scores by rewarding shared medical jargon over diagnostic accuracy. Qualitative audits further exposed hallucinations and spatial attention biases. Ultimately, high semantic scores do not ensure clinical precision, validating the strict necessity of task-specific fine-tuning for autonomous ophthalmic inference.

Index Terms—MedGemma, LOCS III, Ophthalmology, Vision-Language Models, Semantic Similarity, Anterior Segment Imaging Classification.

I. INTRODUCTION

Cataracts remain the leading cause of reversible visual impairment and blindness worldwide, necessitating early and precise diagnosis for optimal surgical planning [1], [2]. The clinical gold standard for standardizing this assessment is the LOCS III, a decimal grading scale that allows ophthalmologists to systematically document cataract progression and anticipate intraoperative challenges during phacoemulsification [3].

Although widely adopted, manual grading is subjective, labor-intensive, and prone to high experience-dependent inter-observer variability [1], [2]. Furthermore, its automation is complicated by imaging artifacts and the challenge of interpreting 3D structures from 2D planes [2], [4], which increases the diagnostic workload and renders large-scale data labeling economically prohibitive [4], [1].

To address these limitations, in general, for medical image analysis, and in particular, for ophthalmological image analysis, are transitioning from traditional convolutional networks to advanced Vision-Language Models (VLMs) and deep-reasoning foundation models [5], [6], [7], [8], [9], [10]. Biomedical foundation models like MedGemma enable zero-shot clinical inference on unannotated scans [11], driving a paradigm shift from simple binary diagnostic labels to cohesive, explanatory prose describing ocular pathologies [5], [11].

However, evaluating generative Artificial Intelligence (AI) introduces a critical “Evaluation Gap.” Traditional lexical metrics (ROUGE-L, BLEU) rely strictly on surface n-gram overlap, failing to capture the true diagnostic intent of clinical reports [12], [13]. In high-stakes medical settings, this necessitates a paradigm shift toward semantic embeddings like BERTScore and sBERT [13], which assess deep contextual similarity to robustly evaluate factual correctness without penalizing valid clinical paraphrasing [12], [13].

This study benchmarks MedGemma against expert ophthalmologists using a private dataset of real-world images. By integrating advanced semantic embeddings with traditional metrics, we rigorously evaluate both categorical LOCS III grading accuracy and the semantic alignment of

generated clinical prose. Ultimately, this work validates the clinical utility of specialized VLMs as safe, interpretable, and effective diagnostic companions.

II. STATE OF THE ART

A. Evolution of Cataract Grading: From LOCS III to Deep Learning

To circumvent the subjectivity and higher inter-observer variability inherent in manual LOCS III grading [14], the field initially pivoted toward traditional Deep Learning methodologies. Early automated approaches primarily leveraged Convolutional Neural Network (CNN)'s such as ResNet and Faster R-CNN [15]. While these classical neural architectures achieved remarkable numerical accuracy in classifying cataracts against LOCS III standards, they were fundamentally constrained by their outputs: opaque, categorical labels devoid of clinical context [16]. This lack of transparency highlighted a critical limitation in traditional Deep Learning, restricting its utility in complex clinical environments that demand nuanced reasoning rather than binary or categorical outputs.

B. Vision-Language Models in Medicine

1) *Architectural Refinements and Specialized Vision Encoders:* To overcome the constraints of narrow classification algorithms, the current technological landscape has seen the emergence of multimodal Foundation Models, such as Med-PaLM M and MedGemma [17], [11]. These VLMs represent a paradigm shift by jointly processing visual inputs and textual data through sophisticated cross-modal alignment [18]. Architecturally, models like MedGemma [11] bypass the limitations of general-purpose visual extractors by integrating specialized medical vision encoders, such as MedSigLIP, which are pre-trained on tens of millions of diverse medical image-text pairs [11]. VLMs can be fine-tuned to a specific medical domain problem and dataset inference [19].

2) *Deep Reasoning and Chain-of-Thought in Multimodal Contexts:* While general-purpose models like GPT-4V and GPT-4o demonstrate high versatility, they occasionally fail to recognize overt pathological findings or struggle with fine-grained spatial interpretation in complex ophthalmic imaging, exposing a critical vulnerability in direct zero-shot clinical inference [18], [20]. To address this, the field is rapidly advancing toward deep reasoning Large Language Model (LLM)s and the implementation of Chain of thought (CoT) prompting methodologies [5]. By forcing the model to decompose its diagnostic process into sequential, logical steps prior to generating a final output, CoT significantly enhances the interpretability of the decision-making pathway [5], [21]. In ophthalmic triage and complex pathological evaluations, structured CoT reasoning partially mitigates the limitations of visual encoders by compensating for visual ambiguities with rigorous, evidence-based textual inference, thereby bridging the gap between raw image processing and expert-level clinical judgment [5], [21].

C. Human-AI Concordance in Clinical Settings

The integration of these highly capable models into ophthalmic workflows necessitates rigorous validation against human expertise. Recent research heavily emphasizes benchmarking AI diagnostic performance against board-certified ophthalmologists to establish a baseline of Expert-level Agreement [20]. Studies evaluating inter-rater reliability have demonstrated that while VLMs can approximate or even match human proficiency in specific text-based reasoning and image-interpretation tasks, their performance fluctuates depending on the complexity of the visual cues [20].

Consequently, the current scientific consensus advocates for the deployment of VLMs strictly as clinical decision-support tools. By functioning as collaborative agents rather than autonomous diagnosticians, these models reduce the cognitive burden of documentation and preliminary triage, while the ultimate diagnostic oversight remains with the senior ophthalmologist [22], [23].

D. The Shift in Evaluation Metrics: From Lexical to Semantic Alignment

As VLMs transition from generating categorical labels to producing complex clinical prose, traditional evaluation methodologies have proven insufficient. Historically, natural language generation tasks relied on lexical metrics such as ROUGE-L and BLEU, which calculate superficial n-gram overlap [12]. However, the literature increasingly critiques these metrics for lacking clinical nuance; they systematically penalize valid medical paraphrasing while occasionally rewarding factually incorrect statements that share surface-level vocabulary [24]. To ensure patient safety, the field is undergoing a critical transition toward embedding-based Semantic Similarity metrics. Leveraging pre-trained biomedical language representations, metrics like BERTScore and sBERT accurately assess whether the diagnostic intent of the AI aligns with the human expert in a continuous latent space, irrespective of the specific syntax utilized [13].

III. METHODOLOGY

A. Dataset Description and Ethical Considerations

The empirical foundation of this study is based on a primary dataset of anterior segment images collected at the Instituto Conde de Valenciana in Mexico City. High-resolution captures were prospectively acquired between September 2024 and January 2025 using the Eyer2 portable anterior segment camera (Phelcom). The study cohort exclusively comprised patients over 40 years of age under clinical evaluation in the glaucoma department, aligning with established research interests in ocular diseases.

To ensure ethical integrity and research confidentiality, the data collection protocol was approved by the Institutional Review Committee (IRC). Written informed consent was obtained from all participants prior to image acquisition, and

all clinical and data handling procedures strictly adhered to the principles of the Declaration of Helsinki.

The final evaluated cohort consisted of 53 patients (one image per patient). Demographic information was available for a subset of 32 patients, comprising 17 males and 15 females. The images were categorically stratified to reflect the clinical variability of real-world practice, resulting in 20 Normal, 27 Cataract, and 6 Pseudophakic cases. In the subset of images positively diagnosed with cataracts, the LOCS III was implemented as the granular labeling standard for Nuclear opalescence (NO), Nuclear Color (NC), and Cortical Opacity (C).

B. Proposed Method and Workflow

The methodological architecture is designed to establish a head-to-head validation between human clinical expertise and algorithmic inference. The workflow operates through two parallel channels:

- 1) **Expert Labeling (Ground Truth Generation):** A panel of certified ophthalmologists evaluated the image cohort. Following the assignment of a primary category, experts graded the severity of cataract cases using the LOCS III scales. Each evaluation concluded with a clinical prose description detailing morphological findings, serving as the definitive ground truth for inter-rater reliability analysis.
- 2) **MedGemma Inference:** Concurrently, the unlabeled images were processed using the MedGemma architecture. A zero-shot clinical inference approach was employed, guided by structured prompt engineering. Without requiring specific fine-tuning on the current cohort, the model was instructed to predict categorical labels, classify the LOCS III grade, and autonomously generate a descriptive clinical report. The model was guided by a structured prompt designed to elicit a systematic anatomical review, as detailed in Fig. 1. This architecture employs a CoT approach, forcing the model to describe ocular structures before arriving at a final diagnosis.

C. Metrics Cross-Validation Strategy and Qualitative Analysis

To evaluate the generated reports, we implemented a dual-metric approach. To establish a baseline for terminological and syntactic precision, we utilized BLEU to measure the co-occurrence of n-grams (incorporating a brevity penalty) [25], alongside ROUGE-L to evaluate structural similarity flexibly through the Longest Common Subsequence (LCS) [26].

To capture underlying diagnostic concepts regardless of surface vocabulary, we employed semantic alignment metrics. BERTScore calculates similarity using dense contextual representations at the token level via greedy matching [27]. Complementarily, sBERT (Sentence-BERT) derives fixed and semantically meaningful embeddings for complete sentences using Siamese network architectures [28], allowing us to

audit the global logical congruence and internal narrative consistency of MedGemma’s reports.

To guarantee the clinical validity of the selected metrics and avoid the algorithmic “illusion of efficacy”, a control experiment based on the cross-validation of diagnostic categories was designed. Specifically, lexical overlap and semantic similarity metrics were calculated by comparing exclusively the texts written by the expert ophthalmologist corresponding to mutually exclusive pathologies, in this case, Cataract vs. Normal, Cataract vs. Pseudophakic, and Normal vs. Pseudophakic. The objective of this methodological triangulation was to establish a threshold or baseline average that revealed how similarity values behave when the underlying clinical state changes drastically. This stress test ensures that metrics such as sBERT or BERTScore are not erroneously rewarding generic syntactic structure or report formatting over the underlying medical diagnosis.

Once the metrics baseline was calibrated, the main experiment was executed by calculating the direct performance between the prose generated by MedGemma [11] and the human specialist’s ground truth. To ensure the analytical rigor of this head-to-head evaluation, a strict methodological restriction was applied: exclusively pairing and computing the text generated by the foundational model and the ophthalmologist’s documentation that unequivocally belonged to the same anatomical image. The results of this metric calculation were projected through standardized graphic representations, allowing for the visualization of performance distribution and the variability of semantic and lexical similarity throughout the entire test corpus. Moreover, we evaluated the ability of MedGemma [11] to correctly classify images according to their diagnostic category (Normal, Cataract, and Pseudophakic). The classification performance was assessed by comparing the predicted diagnostic label extracted from the generated report with the ground-truth label provided by the ophthalmologist.

Finally, to complement the quantitative framework, the evaluation design integrated a qualitative analysis that manually compared the clinical coherence of the text generated by MedGemma [11] with that written by the ophthalmologists. This clinical audit was imperative to detect logical failures and hallucinations that computational metrics in latent space might overlook. The qualitative scrutiny focused on identifying internal diagnostic contradictions within the same report, evaluating the models precision in the morphological identification of synthetic implants (intraocular lenses in pseudophakic patients), and auditing possible spatial attention biases. Critically, algorithmic deviations were monitored to see if the model prioritized describing extrinsic aesthetic attributes or image noise, such as eyelash thickness or corneal reflections, to the detriment of clinically relevant pathological findings in the lens.

IV. RESULTS

The evaluation of MedGemma’s categorical classification capabilities revealed a pronounced disparity between

Experimental Zero-Shot Prompt

This is a photograph of the anterior segment of the eye.

STEP 1 | CLINICAL DESCRIPTION
Describe in detail what you observe in the photograph, following this anatomical order: eyelids and lashes, conjunctiva and sclera, cornea, anterior chamber, iris, lens (including any nuclear opalescence, nuclear color changes, or cortical opacities), and pupil. Write in professional ophthalmic prose, in continuous sentences.

STEP 2 | DIAGNOSIS
Based on your description above, provide your clinical assessment:
a) Overall category | choose one: normal, cataract, pseudophakic, bad quality, needs dilation
b) If cataract, grade the following using the scales provided:

Nuclear opalescence (NO): None/Clear, Very mild, Mild, Mild-moderate, Moderate, Moderate-severe, Severe
Nuclear color (NC): None/Clear, Very mild yellowing, Mild yellowing, Moderate yellow, Yellow-brown, Brown, Dark brown
Cortical opacity (C): None, Peripheral spokes only, Mild peripheral involvement, Moderate spokes approaching center, Central involvement, Severe/dense central spokes

If not cataract, state "not applicable" for the cataract grades.

Fig. 1: Detailed zero-shot chain of thought + prompt architecture designed to enforce anatomical sequence and clinical standardization during the inference process.

identifying healthy anatomy and recognizing pathological states. As illustrated in the confusion matrix (Fig. 2), the model demonstrated an exceptionally high specificity for the “Normal” category, correctly identifying 19 out of 20 cases. However, this high performance masked a critical diagnostic gap; out of 27 confirmed cataract cases, only 5 were correctly classified, while 22 were misclassified as normal. This failure was even more acute in the “Pseudophakic” category, where the model failed to identify a single case, defaulting 100% of instances to the normal class. Table III (Panel A) formalizes these failures with 95% confidence intervals: overall accuracy reached only 0.453 (CI [0.327, 0.585]), and Cohen’s $\kappa=0.100$ (CI [-0.027, 0.239]) encompasses zero, confirming that the observed agreement does not significantly exceed chance. Per-class sensitivity analysis reveals extreme asymmetry: 0.950 (CI [0.751, 0.999]) for *Normal*, but only 0.185 (CI [0.063, 0.381]) for *Cataract* and 0.000 (CI [0.000, 0.459]) for *Pseudophakic*.

Furthermore, evaluating MedGemma’s granular LOCS III sub-grading revealed a systemic inability to stratify cataract severity. Across 27 cataract cases evaluated for NO and NC, the model achieved 0% accuracy, overwhelmingly defaulting to a “None/Clear” prediction in over 70% of instances regardless of the actual clinical severity. While the model seemingly achieved 96% accuracy for Cortical Opacity (C), this was entirely an artifact of the cohort’s distribution; all 27 ground-truth cases presented no cortical opacity. Ultimately, these results confirm a severe algorithmic bias toward baseline

anatomical states, further demonstrating that the model’s high semantic language scores do not translate to granular diagnostic competence.

To analyze the quality of the generated reports, a multidimensional evaluation of the clinical prose was conducted, exposing a fundamental divergence between traditional n-gram matching and contextual embedding scores. As summarized in Table I and visualized across categories in Fig. 3a, lexical metrics were systematically low, reflecting the strict penalty that ROUGE-L (0.213 ± 0.057) and BLEU (0.009 ± 0.006) apply to the varied paraphrasing of generative models. Conversely, semantic metrics demonstrated an upward numerical performance, with BERTScore reaching a mean of 0.845 and sBERT 0.788. The stability of these scores across all categories, including those where the diagnosis was wrong, highlights a “Similarity Paradox” where professional-sounding prose hides clinical incompetence. To rigorously confirm this inflation, Table III (Panel B) provides 95% bootstrap CIs demonstrating that cross-category BERTScore (0.954, CI [0.936, 0.969]) significantly exceeds in-category BERTScore (0.845, CI [0.841, 0.849]) with non-overlapping intervals. Mann-Whitney U tests confirmed this disparity for all four metrics ($p < 10^{-9}$ for BERTScore; Benjamini-Hochberg corrected), with large effect sizes (Cliff’s $\delta > 0.81$), providing definitive statistical evidence that contextual embedding models disproportionately reward shared medical jargon and report structure over diagnostic accuracy.

To isolate the source of this metric inflation, a boilerplate dilution analysis (Table III, Panel C) revealed that 83.6% of the expert reference text consists of shared anatomical templates (e.g., “Lids and lashes normal...”, “Conjunctiva and sclera white...”) common across all diagnostic categories. When only diagnostically relevant segments were isolated—lens-specific phrases such as “Nuclear sclerosis” or “IOL in place”—sBERT similarity dropped sharply from 0.788 to 0.352, confirming that the high full-text scores reflect structural overlap rather than diagnostic accuracy. This quantifies the mechanism underlying the Evaluation Gap: contextual embedding models become clinically blind to the 16.4% of text that carries the actual diagnostic information.

TABLE I: MedGemma performance against the ophthalmologist’s reference diagnosis.

Metric	Mean	Std. Dev.
ROUGE-L	0.213	0.057
BERTScore	0.845	0.014
sBERT	0.788	0.102
BLEU	0.009	0.006

V. CONCLUSION

This study evaluated MedGemma, a biomedical VLM, for anterior segment image classification and clinical report generation. While the model achieved high semantic similarity scores, the cross-validation baseline revealed that these metrics

TABLE II: Fragment of a Qualitative Audit of Clinical Prose: Comparison between Expert Reference and MedGemma Generative Reports.

GT	Pred.	Expert Description (Reference)	MedGemma Clinical Prose	Qualitative Audit (Observations)
cataract	cataract	Superior eyelid ptosis. Conjunctiva and sclera white and quiet without injection or chemosis. Cornea clear with intact epithelium and no stromal haze, edema, keratic precipitates, or neovascularization. Anterior chamber deep and quiet without cells or flare. Iris with normal architecture and no neovascularization or synechiae. Moderate nuclear opacity.	The eyelids appear normal, with eyelashes present. The conjunctiva and sclera are unremarkable. The cornea is clear and smooth, with no visible opacities or significant edema. The anterior chamber depth appears within normal limits, and the anterior chamber angle is not readily discernible in this view. The iris is brown, and the pupil is round and centrally located. The lens is cloudy and opaque, consistent with a cataract. There is no visible nuclear opalescence or cortical opacity. The lens appears to have a yellowish hue.	The model did not detect the ptosis. It is also irrelevant the finding about the eyelashes. The model utilizes non-standard clinical descriptors for corneal morphology. There is a contradiction between “the lens is cloudy and opaque...” and “there is no visible nuclear opalescence ...”.
cataract	normal	Lids and lashes normal in position and contour without blepharitis or lesions. Conjunctiva and sclera white and quiet without injection or chemosis. Cornea clear with intact epithelium and no stromal haze, edema, keratic precipitates, or neovascularization. Peripheral cornea with arcus senilis. Anterior chamber deep and quiet without cells or flare. Iris with normal architecture and no neovascularization or synechiae. Pupil round. Nuclear sclerosis.	The eyelids appear normal, with dark, relatively thick eyelashes present along the upper and lower margins. The conjunctiva and sclera are unremarkable, showing normal vascular patterns. The cornea is clear and smooth, with no visible opacities, edema, or significant irregularities. The anterior chamber depth appears within normal limits, and the iris is visible. The pupil is round and centrally located. The lens is visible, and while it appears generally clear, there is a slight, subtle opacity noted centrally within the lens.	The model generated non-standard clinical descriptions for adnexal structures (eyelashes). The sclera and conjunctiva description is correct with respect to the ground truth. The cornea description is not medically correct (usage of the “smooth” adjective). The model was able to detect a slight case of cataract, but predicted a normal case.

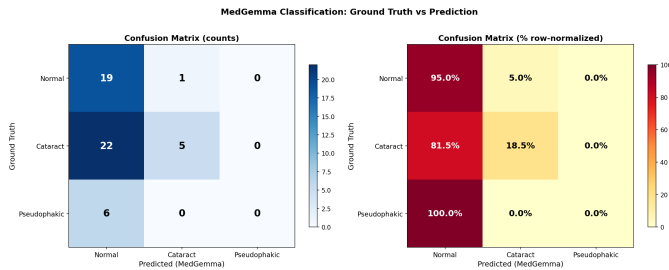
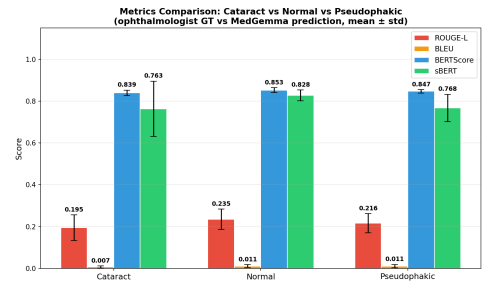


Fig. 2: Confusion matrix: raw counts (left) and row-normalized percentages (right). MedGemma achieves 95.0% specificity for normal cases but shows a 81.5% misclassification rate for cataracts.

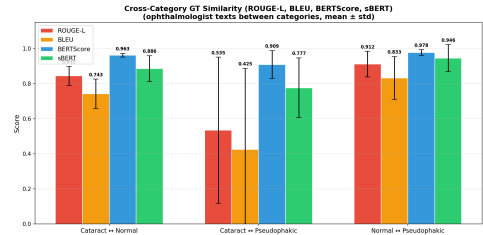
TABLE III: Statistical analysis with 95% confidence intervals. Classification CIs use Clopper-Pearson exact method; text similarity and composite metrics use bootstrap percentile ($B=10,000$).

Analysis	Metric	Value	95% CI
A. Classification Performance			
Global	Accuracy	0.453	[0.327, 0.585]
	Cohen’s κ	0.100	[−0.027, 0.239]
Per-class Sens.	Normal	0.950	[0.751, 0.999]
	Cataract	0.185	[0.063, 0.381]
	Pseudophakic	0.000	[0.000, 0.459]
B. Evaluation Gap (full text)			
BERTScore	In-category	0.845	[0.841, 0.849]
	Cross-category	0.954	[0.936, 0.969]
sBERT	In-category	0.788	[0.758, 0.810]
	Cross-category	0.875	[0.830, 0.916]
C. Boilerplate Dilution (diagnostic segments only)			
sBERT	In-cat (segment)	0.352	—
	Cross-cat (seg.)	0.601	—
	Boilerplate %	83.6%	—

artificially inflate performance by rewarding shared medical terminology and report structure rather than true diagnostic accuracy. Conversely, lexical metrics were substantially lower,



(a) Comparison between lexical metrics (ROUGE-L, BLEU) and semantic embedding metrics (BERTScore, sBERT) across primary diagnostic categories.



(b) Cross-category sensitivity analysis establishing the baseline similarity threshold, identifying the structural influence of medical report templates.

Fig. 3: Quantitative performance analysis of MedGemma’s generative clinical prose against the expert ground truth from the Instituto Conde de Valenciana.

consistent with the limitations of n-gram based evaluation in clinical contexts. Qualitative analysis further identified diagnostic hallucinations, spatial attention biases toward clinically irrelevant artifacts, and internal contradictions in generated reports. These findings demonstrate that high semantic similarity scores do not guarantee clinical precision, and firmly position MedGemma as a clinical decision-support tool requiring expert supervision rather than an autonomous diagnostician.

Despite these insights, the work presents several limitations. The evaluation was conducted on a relatively small dataset collected from a single clinical center, potentially limiting generalizability across different imaging devices, and acquisition protocols. Additionally, experiments were performed in a zero-shot setting; while this assesses out-of-the-box capabilities, it may underestimate improvements achievable through domain adaptation.

Future work should therefore explore few-shot prompting to provide task-specific examples, potentially improving diagnostic grounding without requiring full model retraining. Furthermore, fine-tuning on curated ophthalmic datasets and validating on larger multi-center cohorts will be necessary to assess robustness. Finally, structured evaluations involving multiple ophthalmologists will systematically compare model outputs against inter-expert variability, providing a realistic assessment of clinical reliability.

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